MILITARY SPECIFICATION

SIMULATOR, BOOBY TRAP, FLASH, M117
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly and packing of one type of simulator for the Simulator, Booby Trap, Flash, M117.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-Z-325 - Zinc Plating (Electrodeposited)
QQ-P-416 - Plating, Cadmium (Electrodeposited)

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC 1370

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-286 - Propellants, Solid: Sampling, Inspection and Testing
MIL-STD-810 - Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

8848600 - Simulator, Booby Trap, Flash, M117 Assembly
879971.2 - Box, Packing, Ammunition for Simulator, Booby Trap, Flash, M117, or Simulator, Booby Trap, Illuminating, M118
879973.3 - Box, Packing, Ammunition for Simulator, Booby Trap, Flash, M117, or Simulator, Booby Trap, Illuminating, M118
3139676 - Package, Shipping and Storage (For 25 Simulators, Booby Trap, Flash, M117 with Accessories)
2128179 - Pack, Shipping and Storage (For 50 Simulators, Booby Trap, Flash, M117 with Accessories)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer). ,

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

2.2. Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulations, Title 49, Transportation, Parts 0-190.

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders shall specify 49 CFR 0-190 (latest revision)).

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

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3.2 Assembly. The assembly shall comply with all requirements specified on Drawing (dwg.) 8848600 and with all requirements specified in applicable specifications and standards.

3.3 Moisture content.

3.3.1 Moisture content of match head and flash compositions. The moisture content of the match head composition and the flash composition at the time of loading shall not exceed 0.1 percent when tested as specified in 4.5.1.1 and 4.5.1.2 respectively.

3.3.2 Moisture content of fuze powder. The moisture content of fuze powder, prior to the addition of the cellulose nitrate camphor, at the loading station at the time of loading shall not exceed 0.3 percent when tested as specified in 4.5.1.3.

3.3.3 Felt, paper and chipboard. The moisture content of the felt, paper and chipboard components of the simulator assembly at the loading station, at the time of loading, shall not exceed 6 percent when determined as specified in 4.5.1.4.

3.4 Functioning. The simulator assembly shall ignite and function with a visible flash and produce a sound level intensity of not less that 120 decibels when tested in accordance with 4.5.3.

3.5 Quick leak test. The packed carton sealed in the barrier bag shall show no evidence of air or water leakage when tested as specified in 4.5.4.

3.6 Embrilltlement of extension spring. The extension spring (see dwg. 8848592) shall not break when stretched to the minimum point of permanent distortion, when tested as specified in 4.5.6.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning' agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the items in sample quantities as indicated in Table I.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 6		DRAWING NUMBER
	Simulator, Booby Trap, Flash, M117 Parts For and Loading, Assembling and Packing				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Scratcher Assembly (Dwg. 8848587)				
	Examination for Defects	25		3.2	4.4.2.1
	Extension Spring (Dwg. 8848592)				
	Examination for Defects	25		3.2	4.4.2.2
	Embrittlement Test	10		3.6	4.5.6
	Spool Assembly (Dwg. 8848590)				
	Examination for Defects	25		3.2	4.4.2.3
	Tube Body (Dwg. 8848586)				
	Examination for Defects	25		3.2	4.4.2.4
	Tube Outer (Dwg. 8848595)				
	Examination for Defects	25		3.2	4.4.2.5
	Bracket (Dwg. 8848591)				
	Examination for Defects	25		3.2	4.4.2.6

NOTES:

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 2 OF 6		DRAWING NUMBER
	Simulator, Booby Trap, M117 Parts For and Loading, Assembling and Packing				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Disc (Dwg. 8848581) Examination for Defects	25		3.2	4.4.2.7
	Disc Lower (Dwg. 8848582) Examination for Defects	25		3.2	4.4.2.8
	Disc Upper (Dwg. 8848583) Examination for Defects	25		3.2	4.4.2.9
	Cap (Dwg. 8848588) Examination for Defects	25		3.2	4.4.2.10
	Lower Disc Assembly (Dwg. 8848585) Examination for Defects	25		3.2	4.4.2.11
	Upper Disc Assembly (Dwg. 8848584) Examination for Defects	25		3.2	4.4.2.12

NOTES:

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 3 OF 6		DRAWING NUMBER
	Simulator, Booby Trap, M117 Parts For and Loading, Assembling and Packing				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Flash Assembly (Prior to Assembly in Outer Tube) (Dwg. 8848597)				
	Examination for Defects	25		3.2	4.4.2.13
	Assembly (Prior to Assembling Cap, Loading Flash Charge or Assembling Lower Disc Assembly) (Dwg. 8848600)				
	Examination for Defects	50(a)		3.2	4.4.2.14
	Assembly (After Assembly of Cap but Prior to Assembly of Lower Disc) (Dwg. 8848600)				
	Examination for Defects	50(a)		3.2	4.4.2.15
	Assembly (Dwg. 8848600)				
	Examinations for Defects	50(a)		3.2	4.4.2.16
	Functioning	50(a)		3.4	4.5.3

NOTES:

(a) To minimize sample quantities, these inspections and tests may be performed on the same test items after subsequent assembly operations.

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 4 OF 6		DRAWING NUMBER
	Simulator, Booby Trap, M117 Parts For and Loading, Assembling and Packing				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Unsealed Inner Carton (Dwg. 8799713)				
	Examination for Defects	50		3.2	4.4.2.17
	Unsealed Outer Carton (Dwg. 8799713)				
	Examination for Defects	10(a)		3.2	4.4.2.18
	Sealed Outer Carton (Dwg. 8799713)				
	Examination for Defects	10(a)		3.2	4.4.2.19
	Sealed Packed Barrier Bag (Prior To Placing Into Box (Dwg. 8799713)				
	Examination for Defects	10(a)		3.2	4.4.2.20
	Quick leak test	10(a)		3.5	4.5.4
	Packing Box (Prior to Sealing) (Dwg. 8799712)				
	Examination for Defects	25		3.2	4.4.2.21

NOTES:

(a) To minimize sample quantities, these inspections and tests may be performed on the same test items after subsequent assembly operations.

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 5 OF 6		DRAWING NUMBER
	Simulator, Booby Trap, M117 Parts For and Loading, Assembling and Packing				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Sealed Wooden Packing Box (Dwg. 8799712) Examination for Defects	1 box		3.2	4.4.2.22
	Package, Shipping and Storage (Prior to Closing) (Dwg. 3139676) Examination for Defects	2 boxes (a)		3.2	4.4.2.23
	Package, Shipping and Storage (Dwg. 3139676) Examination for Defects	2 boxes (a)		3.2	4.4.2.24
	Pack, Shipping & Storage (Prior to Closing) (Dwg. 2128179) Examination for Defects	1 box (a)		3.2	4.4.2.25

NOTES:

(a) To minimize sample quantities, these inspections and tests may be performed on the same test items after subsequent assembly operations.

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-S-10521K(AR)

PARAGRAPH CATEGORY	TITLE / EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	DRAWING NUMBER / NEXT HIGHER ASSEMBLY PARAGRAPH REFERENCE /INSPECTION METHOD
	Simulator, Booby Trap, M117 Parts For and Loading, Assembling and Packing		SHEET 6 OF 6		See Below
	Packing, Shipping & Storage (Dwg. 2128179) Examination for Defects	1 box (a)		3.2	4.4.2.26
	Coupler (Dwg. 9321950) Examination for Defects	25		3.2	4.4.2.27
	Extension Cord Packet Assembly (Dwg. 9321948) Examination for Defects	25		3.2	4.4.2.28
	Extension Cord Coupler Assembly (Dwg. 9321949) Examination for Defects	25		3.2	4.4.2.29
	Packing Box, Wirebound (Dwg. 8799713) Examination for Defects	25		3.2	4.4.2.30
	Sealed Wirebound Packing Box (Dwg. 8799713) Examination for Defects	25		3.2	4.4.2.31

NOTES:

(a) To minimize sample quantities, these inspections and tests may be performed on the same test items after subsequent assembly operations.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-S-48078. In addition, inspection lots of simulators shall contain:

a. Parts from one lot interfix from one manufacturer.
b. Composition ingredients from not more than one lot from one supplier.
c. Fuze powder from not more than one lot.
d. Quickmatch from not more than one lot.
e. Match head composition, scratcher composition and charge composition of one or more batches produced under one continuous set of operating conditions, with each batch of the composition having been subjected to the same physical mixing process intended to make the batch homogeneous.

4.4.2 Examination. See MIL-A-48078 (AR). Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Scratcher				8848587
					NEXT HIGHER ASSEMBLY
					8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
Major					
101	Scratcher composition coating missing or unevenly distributed		0.40%	3.2	Visual
102	Scratcher composition not dry		0.40%	3.2	Manual
Minor					
201	Length, min.		0.65%	3.2	Gage
202	Width, min.		0.65%	3.2	Gage
203	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8848592
4.4.2.2	Extension Spring				NEXT HIGHER ASSEMBLY 8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Load at 2 times original length, min.	(a)	0.40%	3.2	Gage
102	Load at 3 times original length, min.		0.40%	3.2	Gage
103	Spring takes permanent set when extended a minimum of 26 inches inside hooks		0.40%	3.2	Manual/Gage
104	Embrittlement test	50	0 - 1	3.6	4.5.6
Minor					
201	Unloaded length		0.65%	3.2	Gage
202	Protective coating missing or contains bare spot		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES: (a) The springs shall be tested in the order listed above.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Spool Assembly				8848590
					NEXT HIGHER ASSEMBLY
					8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Protective coating missing or contains bare spot		0.40%	3.2	Visual
102	Tripwire length	20	0 - 1	3.2	Gage/4.5.7
Minor					
201	Spool split or cracked		0.65%	3.2	Visual
202	Wire insecure		0.65%	3.2	Visual/Manual
203	Protective coating not dry		0.65%	3.2	Visual/Manual
204	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Tube, Body				8848586
					NEXT HIGHER ASSEMBLY
					8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Layers separated or tube damaged to extent that function is impaired		0.40%	3.2	Visual
102	Location of hole		0.40%	3.2	Gage
Minor					
201	Length		0.65%	3.2	Gage
202	Thickness of wall, min.		0.65%	3.2	Gage
203	Inside diameter		0.65	3.2	Gage
204	Crack		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Tube, Outer				8848595
					NEXT HIGHER ASSEMBLY
					8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Layers separated or tube damaged to extent that function is impaired		0.40%	3.2	Visual
102	Vent hole missing, obstructed or mislocated		0.40%	3.2	Visual
Minor					
201	Length		0.65%	3.2	Gage
202	Inside diameter		0.65%	3.2	Gage
203	Wall thickness, min.		0.65%	3.2	Gage
204	Crack		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Bracket				8848591
					NEXT HIGHER ASSEMBLY
					8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness, min.		0.40%	3.2	Gage
102	Mounting hole missing		0.40%	3.2	Visual
Minor					
201	Protective coating missing or contains bare spot		0.65%	3.2	Visual
202	Dimple missing		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Disc				8848581
					NEXT HIGHER ASSEMBLY
					8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Thickness, min.		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8848582
4.4.2.8	Disc, Lower				NEXT HIGHER ASSEMBLY 8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Thickness, min.		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8848583
4.4.2.9	Disc, Upper				NEXT HIGHER ASSEMBLY 8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Thickness, min.		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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Quality Conformance Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8848588
4.4.2.10	Cap				NEXT HIGHER ASSEMBLY 8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Cap torn or punctured		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		1.00%	3.8	Visual

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MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Lower Disc, Assembly				8848585
					NEXT HIGHER ASSEMBLY 8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Lower disc missing or insecure		0.65%	3.2	Visual/Manual
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Upper Disc Assembly				8848584
					NEXT HIGHER ASSEMBLY
					8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Upper disc missing or insecure		0.65%	3.2	Visual/Manual
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Flash Assembly (Prior to Assembly in Outer Tube)				8848597
					NEXT HIGHER ASSEMBLY 8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Body tube ruptured		100%	3.2	Visual
Major					
101	Match head composition missing or length under min.		0.40%	3.2	Visual/Scale
102	Starter paste coating incomplete		0.40%	3.2	Visual
103	Pull cord damaged to extent that function may be impaired		0.40%	3.2	Visual
104	Any component missing, improperly assembled, or insecure		0.40%	3.2	Visual/Manual
105	Scratcher assembly and igniter pad improperly assembled (not permitting free passage of pull cord)		0.40%	3.2	Visual
106	Quickmatch insecure		0.40%	3.2	Visual/Manual
107	Upper disc assembly insecure		0.40%	3.2	Manual
Minor					
201	Knot missing from uncoated end of pullcord		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

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Quality Conformance Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Assembly (Prior to Assembly Cap, Loading Flash Charge or Assembling Lower Disc Assembly)				8848600
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Pull cord insecurely taped as evidence by a slight pull parallel to longitudinal axis of assembly		100%	3.2	Visual/Manual
Major					
101	Any component missing		0.40%	3.2	Visual
102	Glue on pull cord		0.40%	3.2	Visual
Minor					
201	Part or sub-assembly can be removed by applying a slight finger pressure		0.65%	3.2	Manual
202	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Assembly (After Assembly of Cap but Prior to Assembly of Lower Disc)				8848600
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Flash charge missing		0.40%	3.2	Visual
102	Evidence of flash composition in outer tube		0.40%	3.2	Visual
103	Cap improperly assembled or insecurely taped		0.40%	3.2	Visual/Manual
104	Outer tube vent hole not sealed by the sealing tape (applies after assembling cap & following 24 hrs drying period)		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8848600
4.4.2.16	Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Sufficient paint leakage under cap to bond pull cord to cap (inspection for the presence of this defect need only be performed on those assemblies upon which the tape is insecure, applied less than 360 degrees, or has wrinkles extending from tape edge to beyond cap and outer tube joint)		100%	3.2	Visual
Major					
101	Bracket missing, damaged or insecure		0.40%	3.2	Visual/Manual
102	Lower disc assembly insecure		0.40%	3.2	Manual
103	Outer tube vent hole covered by bracket		0.40%	3.2	Visual
104	Vent hole sealing tape not completely covered by binding tape		0.40%	3.2	Visual
105	Functioning	4.4.3.3		3.4	4.5.3
Minor					
201	Any component missing		0.65%	3.2	Visual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Bare spot in painted surface		0.65%	3.2	Visual
204	Tape pull tab missing or inadequate		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8799713
4.4.2.17	Unsealed Inner Carton				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Simulator missing or damaged		0.40%	3.2	Visual
102	Any accessory component missing or damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.18	Unsealed Outer Carton				8799713
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inner carton missing or improperly sealed		0.40%	3.2	Visual
102	Inner carton damaged to extent that contents are exposed		0.40%	3.2	Visual
103	Instruction sheet missing		0.40%	3.2	Visual
104	Carton, filler or simulator obviously damp or wet		0.40%	3.2	Visual/Manual
Minor					
201	Marking misleading or undefinable		0.40%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Sealed Outer Carton				8799713
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Carton damaged to extent that contents are exposed		0.40%	3.2	Visual
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Marking misleading or undefinable		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.20	Sealed Packed Barrier Bag (Prior To Packing Into Box)				8799713
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Waterproofness of barrier bag destroyed by rupture, puncture, or separation of heat seal	(a)	0.40%	3.2	Visual
102	Evidence of crushed carton		0.40%	3.2	Visual
Minor					
201	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied		0.65%	3.2	Visual
202	Marking of bag misleading or undenfinable		0.65%	3.2	Visual

NOTES:

(a) Package should be opened for inspection if there is evidence of crushing.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8799712
4.4.2.21	Packing Box (Prior to Sealing)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Packed barrier bag missing		0.40%	3.2	Visual
102	Packing component missing		0.40%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8799712
4.4.2.22	Sealed Wooden Packing Box				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged to extent that contents are exposed or liable to become exposed		0.40%	3.2	Visual
102	Strapping missing, broken or loose		0.40%	3.2	Visual/Manual
103	DOD symbol misleading or undentifiable		0.40%	3.2	Visual
Minor					
201	Strapping mislocated or improperly engaged		0.65%	3.2	Visual
202	Contents loose		0.65%	3.2	Manual
203	Marking misleading or undentifiable		0.65%	3.2	Visual
204	Metallic seal missing, unsealed or incorrectly positioned		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.23	Package, Shipping and Storage (Prior to Closing)				3139676
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any packing component missing or misassembled		0.40%	3.2	Visual
Minor					
201	Number of desiccant units incorrect		0.65%	3.2	Visual
202	Desiccant bags not properly secured		0.65%	3.2	Visual
203	Indicator card missing		0.65%	3.2	Visual
204	Number of inner packages incorrect		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.24	Package, Shipping and Storage				3139676
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	CFR markings missing or unidentifiable		100%	3.2	Visual
Major					
101	Tray interferes with seal of lid		0.40%	3.2	Visual/Manual
Minor					
201	Descriptive marking missing, misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.25	Pack, Shipping and Storage (Prior to Closing)				2128179
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Package missing or improperly positioned		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 2128179
4.4.2.26	Pack, Shipping and Storage				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	CFR marking missing or undentifiable		100%	3.2	Visual
Major					
101	Wire binding not properly secured		0.40%	3.2	Visual
102	Metallic seal missing		0.40%	3.2	Visual
Minor					
201	Descriptive marking missing, misleading or undentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.27	Coupler				9321950
					NEXT HIGHER ASSEMBLY
					8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Length		0.65%	3.2	Scale
202	Width		0.65%	3.2	Scale
203	Thickness		0.65%	3.2	Gage
204	Diameter and location of small hole		0.65%	3.2	Gage
205	Contour of keyhole		0.65%	3.2	Visual
206	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9321948
4.4.2.28	Extension Cord Packet Assembly				NEXT HIGHER ASSEMBLY 8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length of extension cord		0.40%	3.2	Scale
Minor					
201	Length of loops		0.65%	3.2	Scale
202	Width of loops		0.65%	3.2	Scale
203	Cord insecurely taped		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9321949
4.4.2.29	Extension Cord Coupler Assembly				NEXT HIGHER ASSEMBLY 8848600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Coupler missing or insecurely attached		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.30	Packing Box, Wirebound, Prior to Sealing (Alternative)				8799713
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Barrier bag missing or damaged		0.40%	3.2	Visual
102	Packing component (filler) missing or improperly positioned		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-10521K(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8799713
4.4.2.31	Sealed Wirebound Packing Box (Alternative)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged to extent that contents are exposed		0.40%	3.2	Visual
102	Wire binding missing, broken or loose		0.40%	3.2	Visual
Minor					
201	Wire binding mislocated or improperly engaged		0.65%	3.2	Visual
201	Contents loose		0.65%	3.2	Manual
203	Marking misleading, unsealed or unidentifiable		0.65%	3.2	Visual
204	Metallic seal missing, unsealed or incorrectly positioned		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.3.1 Moisture content (see Table II) - Major defect

TABLE II - Moisture Content

Material
Match head composition (see 3.3.1)
Flash composition (see 3.3.1)
Fuze powder (see 3.3.2)
Felt (see 3.3.3)
Paper (see 3.3.3)
Chipboard (see 3.3.3)
Carton and carton packing material (see dwg. 8799713)

The contractor shall provide adequate controls to insure that the materials comply with the requirements. For verification, the contractor shall select and subject to test, one sample of each material from each 8 hours production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirement, that quantity of sub-lot of material represented by the sample shall not be used in production. If the quantity of material or sub-lot with excessive moisture has been used in loading and packing, the remaining unloaded and unpacked material shall not be used in production, and the loaded and packed simulators shall be rejected. Test shall be performed as specified in 4.5.1.

4.4.3.2 Determination of match head, scratcher and flash composition - Major defect - (see Table III). A sample quanitity of each composition shall be selected from each batch for these determinations. If the sample fails to meet the requirements specified, the batch shall be rejected. Test shall be performed as specified in 4.5.2.

TABLE III - Composition Content

Composition
Match head (see dwg. 9269240)
Scratcher (see dwg. 9269237)
Flash (see dwg. 9269238)

4.4.3.3 Functioning. The simulator assemblies shall be observed for any evidence of failure to comply with the requirements as classified in Table IV, when tested as specified in 4.5.3.1.

TABLE IV - Functioning Requirement

Defect	Classification
Dud (see 3.4)	Major
No visible flash (see 3.4)	Major
Sound level below minimum (see 3.4)	Major
Spring breaks or fails to function (see 3.6)	Major

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4.4.3.3.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the applicable requirements specified, one hundred and twenty-five (125) simulator assemblies shall be selected from each lot for test. The lot shall be rejected if, during the test, six (6) or more assemblies exhibit any of the major defects as listed in Table IV.

4.4.3.3.2 After three consecutive lots have complied with the acceptance criteria of 4.4.3.3.1, fifty (50) assemblies shall be selected from each lot for test. The lot shall be rejected if, during the test, three (3) or more assemblies exhibit any of the defects as listed in Table IV.

4.4.3.4 Quick leak test. Major defect - Ten (10) cartons in sealed barrier bags shall be selected from each lot for this test. If two (2) or more packages show evidence of air leakage during test or water leakage when package is opened, the lot shall be rejected. The test shall be performed as specified in 4.5.4.

4.4.3.5 Protective coating test of extension spring (dwg. 8848592) and Bracket (dwg. 8848591) - Major defects. Sampling, acceptance and rejection shall be in accordance with QQ-Z-325 or QQ-P-416 whichever is applicable except that actual parts may be used in lieu of specimens. The test procedure shall be as specified in 4.5.5.

4.4.3.6 Improper seating of lower disc assembly and flash powder leakage - Major defects. One hundred twenty five (125) simulators shall be randomly selected from each lot. The simulators shall be vibrated and radiographically inspected for improper seating of the lower disc assembly within the body tube and presence of loose flash powder in the outer tube. The lot shall be rejected if one or more of the major defects are found. The test shall be performed as specified in 4.5.8.

4.4.4 Inspection Equipment. Inspection equipment required to perform the examinations and tests prescribed in herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6. herein.

4.5 Methods of inspection.

4.5.1 Moisture content.

4.5.1.1 Match head composition. An accurately weighed portion of approximately 5 grams (gm) of composition, prior to the addition of the lacquer, shall be transferred to a tared aluminum or glass weighing dish approximately 60 millimeters (mm) in diameter. A

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tight fitting cover shall be fitted to the dish. The cover shall be removed, and-the dish and contents placed in a desiccator containing a suitable desiccant such as calcium chloride, calcium sulfate or sulfuric acid. The desiccator shall be maintained at a temperature of 25 + 5°C, 77 ± 10%. The dish and contents shall remain in the desiccator until the loss in weight between subsequent weighings does not exceed 1 milligram. The loss in weight shall be calculated as follows:

Percent moisture = 100A divided by W

where:
A = loss in weight in grams.
w = weight of the sample in grams.

4.5.1.2 Flash composition. An accurately weighed portion of approximately 2 gms of flash composition shall be transferred to a tared moisture dish. The moisture dish and contents shall be placed in a drying oven at 85 ± 5°C, 185 ± 10°,F for three (3) hours, cooled in a desiccator and weighed. The percentage of moisture in the sample shall be calculated as follows:

Percentage of moisture = 100A divided by W

where:
A = loss in weight in grams.
w = weight of the sample in grams.

4.5.1.3 Fuze powder. Prior to adding the cellulose nitrate camphor binder, if used, an accurately weighed portion of approximately two (2) gms of the fuze powder shall be transferred to a tared weighing dish or watch glass. The container and contents shall be dried for four (4) hours in an oven at 70 ± 5°C, 158 + 10°F. The dish and contents shall then be removed, covered, cooled in a desiccator and weighed. The loss in weight shall be calculated to the percentage of moisture in the sample.

4.5.1.4 Felt, paper, chipboard, carton and carton packing material. Ten (10) gms of each material shall be accurately weighed and placed in a tared weighing dish. Each dish and contents shall be weighed and placed in an oven and dried at 100 ± 2°C, 212 + 4%?, for 2 hours. Each dish and contents shall be cooled in ± desiccator and weighed. The loss in weight shall be calculated as percent moisture in each sample.

4.5.1.5 Alternate method. The moisture content shall be determined in accordance with Method T101.4 of MIL-STD-286. The timer shall be set to the minimum time required to obtain constant readings and the drying temperatures shall be as specified in 4.5.1.1 through 4.5.1.4

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4.5.2 Determination of compositions.

4.5.2.1 Match head. Prior to adding the lacquer, a one gram sample shall be selected and analyzed as follows:

4.5.2.1.1 Antimony sulfide. An accurately weighed portion of approximately 1 gm shall be transferred to a 150 milliliter (ml) beaker . Twenty-five ml's of 12N hydrochloric acid shall be added to the beaker, heated on a steam bath for 30 minutes with occasional stirring, and then filtered through a tared medium porosity glass crucible with the aid of suction. The beaker and contents of the crucible shall be washed with a two percent solution of hydrochloric acid and finally with four 5 ml. portions of water. The crucible and contents shall be dried in an oven maintained at 100 + 5°C, 212 ± 10°F, for one hour, cooled in a desiccator for 15 minutes and weighed. The percentage of antimony sulfide in the sample shall be calculated as follows:

Percentage of antimony sulfide = A - [(B-C) + (DE) x 100 divided by A]

where:

A = weight of sample
B = weight of crucible and residue.
c = weight of crucible.
D = loss in weight of crucible and contents after extraction with 15N nitric acid as determined in 4.5.2.1.2.
E = factor required by volubility of red phosphorus in hydrochloric acid (see 4.5.2.1.1.1).

4.5.2.1,1,1 Calculation of factor E. From the red phosphorus used in the manufacture of the match head composition, 0.5300 plus or minus 0.001,0 gm. shall be transferred to a 150 ml. beaker and 25 ml. of 12N hydrochloric acid added. The beaker and contents shall be heated on a steam bath for 30 minutes, stirred occasionally, and then filtered through a tared, medium porosity glass crucible, with the aid of suction. The beaker and contents of the crucible shall be washed with a two percent solution of hydrochloric acid and finally with four 5 ml portions of water. The crucible and contents shall be dried in an oven mainatained at 100 + 5°C, 212 + 10°F, for one hour, cooled in a desiccator for 15 minutes and weighed. The loss in weight of the crucible and contents shall be calculated to percentage of red phosphorus soluble in 12N hydrochloric acid. The factor represented by the letter E in the equation for percentage of antimony sulfide shall be calculated as follows:

E = F divided by G

where:

F = percent volubility of the red phosphorus used in the composition, expressed as a decimal fraction.
G = percent of volubility of the red phosphorus subtracted from 100 and the results expressed as a decimal fraction.

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4.5.2.1.1.2 In the course of analysis of samples of match head composition, the volubility in hydrochloric acid of red phosphorus shall be recorded whenever practicable.

4.5.2.1.2 Red phosphorus. The residue obtained in 4.5.2.1.1 shall be transferred from the glass crucible to a 150 ml. glass beaker with the aid of a stirring rod and five 5-ml. portions of 15N nitric acid. The crucible shall be rinsed with four 5-ml. portions of distilled water, and the rinsing caught in the beaker. The contents of the beaker shall be heated on a steam bath for one hour, and filtered with the aid of the previously used crucible using suction. The beaker and contents shall be washed with four 5-ml. portions of distilled water. The crucible and residue shall be dried in an oven maintained at 100 ± 5°C, 212 ± 10°F, for one hour, cooled in a desiccator for 15 minutes, and weighed. The percentage of red phosphorus in the sample shall be calculated as follows:

Percentage of red phosphorus = 100D divided by F A

where:

D = loss in weight of crucible and contents due to extraction with 15N nitric acid.
F = factor obtained by subtracting the percent volubility of the red phosphorus from 100 and expressing the results as a decimal fraction (see 4.5.2.1.1.1).
A = weight of sample.

4.5.2.1.3 Charcoal. percentage of charcoal in the sample shall be calculated as follows:

Percentage of charcoal = G-C divided by A x 100

where:

G = weight of crucible and charcoal.
c = weight of crucible
A = weight of sample.

4.5.2.2 Scratcher composition. The following analysis shall be performed prior to adding the water to the composition.

4.5.2.3 Antimony sulfide. An accurately weighed portion of approximately one gm of the sample which had been previously dried at 100 ± 5°C, 212 ± 10°F shall be transferred to a 150 ml. beaker. Fifty (50) ml. of water shall be added to the beaker, the contents heated on a steam bath for one hour, and then filtered using a tared, Gooch crucible. The beaker and residue shall be washed five (5) times with hot water, and the residue caught in the above mentioned crucible. The crucible and contents shall be dried for one hour at 100 to 105°C, 212 to 222°F cooled in a desiccator and weighed. The filtrate shall be retained for the determination of potassium chlorate and dextrin. The gain in weight of the crucible shall be calculated to percentage of antimony sulfide in the sample.

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4. 5.2.2.2 Potassium chlorate. The filtrate obtained in 4.5.2.2.1 shall be diluted to 250 ml. in an accurately calibrated volumetric flask. A 25 ml. aliquot shall be transferred with a calibrated pipette to a 500 ml. Ehrlenmeyer flask, diluted to approximately 150 ml. and approximately 50 ml. of 18N sulfuric acid added. The flask shall be stoppered with a rubber stopper equipped with a Bunsen valve and the contents heated almost to boiling. Fifty (50) ml. of a standardized solution of approximately four percent ferrous ammonium sulfate (Mohr's Salt) shall be added, using a calibrated pipette, and the solution boiled for two minutes, cooled, the valve loosened, and the excess of Mohr's Salt titrated with a standardized 0.1N solution of potassium permanganate. Currently conduct a blank determination using the same quantities of the same reagents used for the analysis of the sample. The percentage of potassium chlorate in the sample shall be *calculated as follows:

$$\text{Percent of potassium chlorate} = .2043\ (V-v)\ N100 \text{ divided by } W$$

where:

V = ml. of standard potassium permanganate required for blank titration, ml.
v = volumne of potassium permanganate required for sample titration, ml.
N = normality of potassium permanganate solution.
w = weight of original sample, grams.

4.5.2.2.3 Dextrin. The percentage of dextrin in the sample shall be calculated by subtracting from 100 the combined percentages of antimony sulfide and potassium chlorate.

4.5.2.3 Flash composition.

4.5.2.3.1 Magnesium. An accurately weighed portion of approximately one gram of the sample shall be transferred to a 150 ml. beaker . One hundred (100) ml. of water at 25 ± 3°C, 77 + 10°F, shall be added to the beaker. The contents of the beaker shall be stirred with a stirring rod for two or three minutes to dissolve the potassium perchlorate, and the contents filtered by means of a tared, 30 ml. medium porosity, fritted-glass crucible. (It is essential that the filtration be carried out immediately after the completion of the two to three minute stirring period, otherwise appreciable reaction will result which will adversely affect the accuracy of the results obtained). The residue shall be transferred to the crucible quantitatively and washed with five 5-ml. portions of water, and two 5-ml. portions of acetone. The crucible and contents shall be aspirated for two minutes, dried for 30 minutes at 100 ± 5°C., 212 ± 10°F, cooled in a desiccator, and weighed. As much of the residue remaining in the crucible as practicable, shall be transferred to a 150 ml. beaker. One hundred

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(100) ml. of five percent acetic acid solution shall be added to the beaker and the contents stirred occasionally until there is no longer a visible reaction. Twenty (20) ml. of five percent acetic acid solution shall be added to the crucible and the contents stirred occasionally until there is no longer a visible reaction. The contents of the beaker shall then be filtered using the same filtering crucible mentioned above. The residue in the beaker shall be transferred to the crucible quantitatively and washed with five 5-ml. portions of water and two 5 ml. portions of acetone. The crucible and contents shall be dried for 30 minutes at 100 ± 5° C, 212 ± 10°F, cooled in a desiccator, and weighed.

The loss in weight, due to the five percent acetic acid treatment, shall be calculated to percent magnesium in the sample on a dry basis.

4.5.2.3.2 Antimony sulfide. The residue remaining in the crucible, obtained in connection with the determination of magnesium shall be calculated to percent antimony sulfide in the sample of a dry basis.

4.5.2.3.3 Potassium perchlorate. The percent potassium perchlorate in the sample shall be calculated by subtracting from 100 the combined percentage of magnesium and antimony sulfide.

4.5.3 Functioning.

4.5.3.1 Functioning. The simulator assemblies shall be mounted in accordance with the instruction drawing 8848620. The assemblies shall be ignited by cutting the trip wire. The sound level meter or microphone for recording sound level intensity shall be placed a . distance of 25 ± .5 feet from the simulator in the direction of the wi nd. Each assembly shall be observed for compliance with the requirements of 3.4. Test equipment shall be in accordance with 4.4.4. (Destructive Test)

4.5.3.2 Test validity. If for any reason the testing agency considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and-a new test performed with addtional samples.

4.5.4 Quick leak test (Vacuum Differential Method). The filled and sealed packages, after conditioning at ambient temperature (21 ± 2°C, 70 + 5°F) for at least four (4) hours, shall be tested for leakage by submerging in water contained in a vacuum vessel. A vacuum differential of 216mm (8 1/2 inches) of mercury from ambient conditions shall be drawn and held for 30 seconds (minimum) while observing for leakage of air. Repeat the test after inverting the sample. A steady stream or recurring succession of bubbles from any surface or seam shall be cause for rejection. Where there is no emission of bubbles observed, the package shall be dried, opened and inspected. Evidence of water within the barrier bag, or on the carton, shall also be cause for rejection. (Non-Destructive Test)

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4.5.5 Protective coating of extension spring and bracket. The adequacy of the protective coating shall be tested in accordance with the procedures specified in QQ-Z-325 and QQ-P-416 whichever is applicable.

4.5.6 Embrittlement of extension spring. The spring shall be stretched along its longitudinal axis until it no longer returns to its original position. When the spring no longer returns to its original length or profile it shall be considered permanently distorted. Any spring that breaks during the test shall be classed defective. (Destructive Test)

4.5.7 Length of trip wire. The trip wire shall be unwound from the spool assembly and its length measured to determine compliance with the specified requirement. (Non-Destructive Test)

4.5.8 Improper seatin og lower disc assembly and flash powder leakage. The simulators shall be vibrated in accordance with the equipment and procedures specified in MIL-STD-810, Method 514.1, shipment by common carrier land or air, Figure 514.1-7, Curves AX and AY, Table 514.1-VII. Subsequent to vibration, the items shall be radiographically inspected and compared with radiographic standards for detection of defects. The contractor shall prepare radiographic standards showing the following conditions:

a. One (1) acceptable item (flash tube appears centered in outer tube, i.e. center lines of flash tube and outer tube in line) snowing disc completely seated within the flash tube and no evidence of powder in outer tube;

b. One (1) acceptable item (flash tube appears off center in outer tube, i.e., 900 rotation from "a" above) showing disc completely seated within the flash tube and no evidence" of powder in outer tube;

c. One (1) unacceptable item showing disc completely out of phase with flash tube and evidence of powder in outer tube;

d. One (1) unacceptable item showing disc not completely seated within flash tube and evidence of powder in outer tube.

The above standards and the radiographic technique used shall be submitted to ARRADCOM, DRDAR-QAT-M, Dover, NJ 07801 for approval prior to use for acceptance inspection. The sample items subjected to radiographic analysis shall not be submitted for ballistic test. (Destructive Test)

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. The simulator assemblies shall be preserved and packaged in accordance with dwg. 8799713.

5.1.2 Level A. For Marine Corps use, preservation and packaging shall be in accordance with dwg. 3139676.

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5.2 Packing.

5.2.1 Level A. The cartons shall be packed in accordance with dwg. 8799712.

5.2.2 Level A. For Marine Corps use, packing shall be in accordance with dwg. 2128179.

5.3 Marking.

5.3.1 The boxes shall be marked as specified on dwgs. 8799712 and 8799713.

5.3.2 For Marine Corps use, marking shall be as specified on dwgs. 2128179 and 3139676.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the M117 Booby Trap, Flash Simulator.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples. Provisions of MIL-A-48078.
d. For Marine Corps Packing, and Marking see section 5.1.2, 5.2.2 and 5.3.2.

6.2 Submission of inspection equipment designs for approval. (See MIL-A-48078). Submit equipment designs, as required, to: Commander, U.S. Army Armament Research & Development Command, ATTN: DRDAR-QAT-I, Dover, New Jersey 07801.

6.3 Distribution of ammunition data cards. Distribution of data cards shall include the following: Commander, US. Army Armament Research & Development Command, ATTN: DRDAR-QAT-M, Dover, New Jersey 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading, "US Army Armament Research and Development Command (ARRADCOM)" may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsental, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Changes from Previous Issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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